Problem no:21

Problem Name : Quantum Protocol & Algorithm Visual Simulator

Team Details

Team Name :
Department Name : CTech
Campus Name : SRM Kattankulanthur
Team Size : 2
Team member 1 name : Subhashree Satpathy
Team member 2 name : Serina Mayanglambam
Team member 3 name :
Team member 4 name :

Problem Statement

Problem Description

Quantum simulators allow researchers and students to explore quantum systems without requiring real quantum hardware. The goal of this project is to design an interactive visual simulator that demonstrates how quantum protocols and algorithms operate.

The simulator will visualize quantum state evolution, superposition, and measurement probabilities, enabling users to observe how quantum gates manipulate qubit states.

Tasks

- Simulate qubit state evolution
- Implement basic quantum gates and circuits
- Visualize quantum state changes using Bloch sphere representation
- Demonstrate simple quantum algorithms

Expected Outcomes

- Interactive Bloch sphere animation
- Quantum circuit visualization
- Measurement probability plots
- Step-by-step algorithm execution

Key Concepts

Qubits, Superposition, Quantum Gates, Measurement , Quantum Algorithms

Objectives &Key Concepts

Objectives

- Develop an interactive quantum simulator
- Demonstrate quantum state evolution visually
- Allow users to experiment with quantum circuits
- Improve conceptual understanding of quantum computing

Key Concepts

- Quantum Superposition – A qubit can exist in a combination of $|0\rangle$and $|1\rangle$.
- Quantum Gates – Operations like Hadamard, Pauli-X, and CNOT modify qubit states.
- State Vector Representation – Describes the quantum state using probability amplitudes.
- Measurement Probabilities – Measurement collapses the qubit to $|0\rangle$ or $|1\rangle$ based on probabilities.
- Bloch Sphere Visualization – A 3D representation of qubit states and their transformations.

Theory

- A **qubit** is the basic unit of quantum information and can exist in **superposition** of $|0\rangle$and $|1\rangle$.

  A general qubit state is represented as

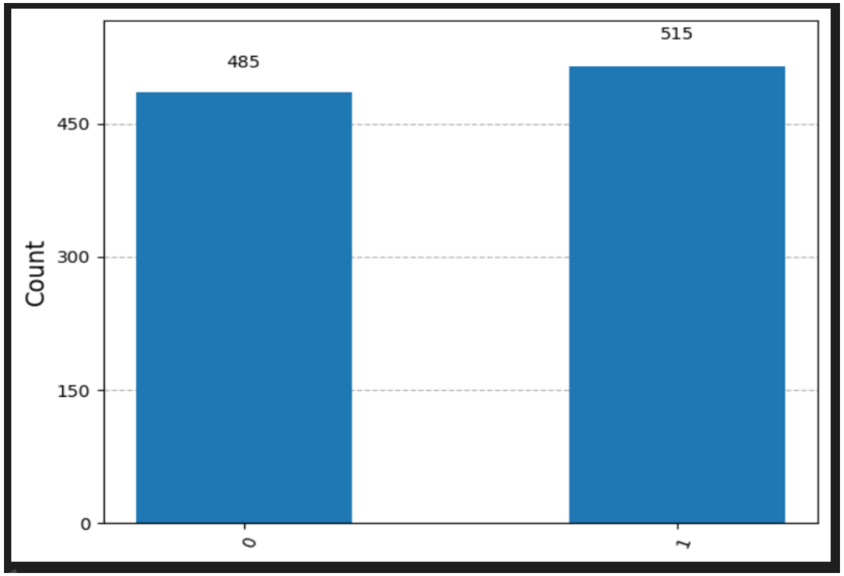
  $$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

  α and β are probability amplitudes where $|\alpha|^2 + |\beta|^2 = 1$.
- **Quantum gates** perform unitary operations that change the qubit state.
- The **Bloch sphere** provides a 3D visualization of qubit state evolution.
- **Quantum simulators** help visualize quantum circuits, superposition, and measurement probabilities without real hardware.

Simulation

Hadamard Gate Simulation

- The Hadamard gate places the qubit into superposition, creating an equal combination of $|0\rangle$ and $|1\rangle$.
- Measurement results in simulation show approximately 50% $|0\rangle$ and 50% $|1\rangle$ probabilities.



Grover's Algorithm Simulation

- Grover's algorithm searches for a target state by using an oracle and diffusion operator.
- The algorithm amplifies the probability of the correct state, making it most likely to appear during measurement.

Animation

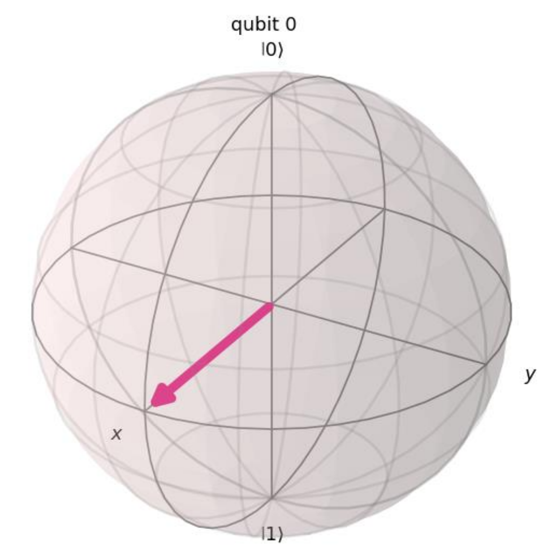
Handamard Gate

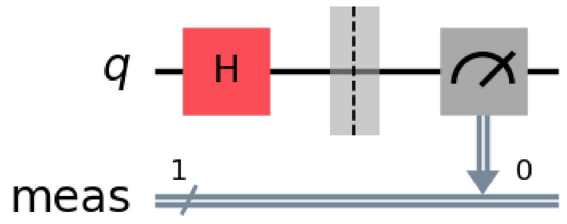
Bloch Sphere (Quantum State Evolution)

- The Bloch sphere represents a qubit state in 3D.
- The state vector shows amplitudes of $|0\rangle$and $|1\rangle$.
- A Hadamard gate rotates the state vector.
- The qubit changes from $|0\rangle$ to superposition.

Quantum Circuit Visualization

- Quantum circuits apply operations to qubits.
- Quantum gates modify the qubit state.
- The Hadamard gate creates superposition.
- Measurement collapses the state to 0 or 1.

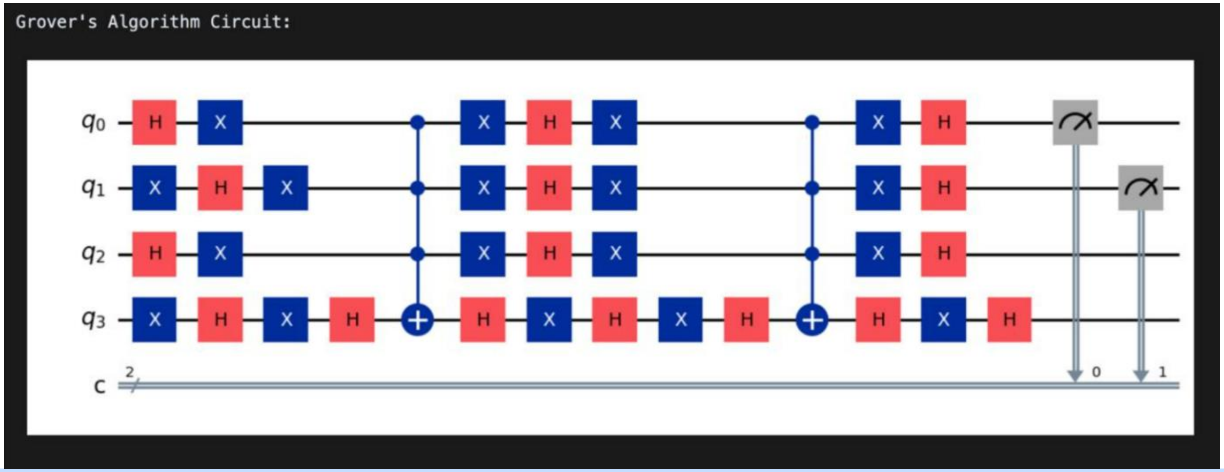




Animation

Grover's Algorithm Circuit

- Grover's algorithm is used for searching an unsorted database.
- Hadamard gates create superposition across all states.
- The Oracle marks the target state.
- The Diffusion operator amplifies the probability of the correct state.
- Measurement returns the desired solution with high probability.
- Provides quantum speedup: $O(\sqrt{N})$vs classical $O(N)$.



Conclusion

- An **interactive quantum visual simulator** was developed to demonstrate qubit behavior and fundamental quantum operations.
- The simulation shows how the **Hadamard gate creates superposition**, producing nearly equal probabilities of measuring $|0\rangle$ and $|1\rangle$.
- The implementation of **Grover's algorithm demonstrates quantum search using amplitude amplification** to increase the probability of the target state.
- Visualization tools such as the **Bloch sphere, quantum circuits, and probability histograms** help in understanding quantum state evolution.

Future Work

- Extend the simulator to support **multi-qubit quantum systems**.
- Implement **quantum entanglement simulations** to demonstrate correlations between qubits.
- Add **interactive controls** such as sliders for selecting quantum gates and parameters.
- Simulate additional quantum algorithms such as **Deutsch–Jozsa or Shor's algorithm**.
- Enhance visualization with **real-time animations and interactive quantum circuit builders**.

Thank You